How We Tested

Testing TV tuners is a very subjective process, as we try to compare the quality of captured content amongst all the contenders. Some aspects of TV tuners—such as MPEG-II encoding in real-time—are hardware specific, but this process also hogs CPU time. Video encoding takes a heavy toll on system resources, irrespective of your configuration.

The test bench consisted of a Pentium 4, 2 GHz, on an Intel D850EMV motherboard, with 256 MB of RD RAM. The video card was a GeForce4 440MX, and the sound card used was a Creative 1373. We tested the cards to capture content using both, cable and S-Video, as the input sources. We used Ulead Media Studio to capture content from the S-Video input in raw format-RGB (24-bit) with CD quality audio (wav files). This was done to check the quality of the captured content without any changes or compression, made by either the hardware or software. We also used the cards to capture regular cable content using the applications and software supplied by the manufacturers. This was done to check both, the quality of content, and also the real-world performance of the card. Any issues such as crashing of applications and drivers, or dropping of frames were duly noted.

A restore point of a fresh installation of Windows XP, with all the necessary drivers and patches installed, was made. The system was restored to this point each time we installed a new card, to avoid any possible driver and software conflicts.

Needless to say, the cards performed as they were meant to, however it was the features and the final content quality that separated the good from the mediocre.

Coming to the software aspect of this month's tests, the PVR (Personal Video Recording) software test looked for noticeable differences or advantages over bundled software. As all PVR software rely on the same hardware device—the TV tuner card—they can only be adjudged on the basis of the features they offer. However, we noted a software's effort towards improving the quality of captured images in terms of noise reduction, video smoothing, etc.

We looked to answer all the questions you ask when buying PVR software: Can I play or pause a Live TV signal? Does it have the Instant Replay function? Can I skip the commercials, and catch up with the live TV signal? How long can the instant replay be? In what formats can I store the captured video?

We also tested each software for ease of use, to answer a few more questions you might have: How much time will I spend configuring the software? How easy it is to locate and set parameters? How good is the documentation? How good are the software's default settings?

bundled its own software. Any feature such as picture improvisation or NICAM stereo, would have definitely given this little card a boost in the scores here.

Performance

Performance finally decided which card was the best. We captured content in MPEG-2, for the cards that supported it, and in MPEG-1 for the cards that didn't. This was done to test a card's capabilities, including dropped frames and processor usage. Processor usage is a factor that needs attention, because a computer with low resources may not be able to directly capture data in the MPEG-2 format. A Pentium III 700 MHz processor is good enough, but for MPEG-2 recording, a Pentium III 1 GHz system is recommended. Another important aspect is the cable reception signal—often a weak signal results in considerable frames being dropped when capturing data in the RAW format (RGB 16).

Other factors also come into play *(See box, 'Heads-up' on page 55)*. The software bundled with tuner cards is very utilitarian—you can do PVR work, as well as minor video editing, without having to pay more for software. All the cards captured content at the highest resolution that PAL supports. A cable signal is transmitted at a resolution of 720 x 576, and if content is captured at 640 x 480, it looks decent enough. The amount of dropped frames count, and the overall captured content quality may not be impressive, but a little tweaking will definitely get you satisfactory results.

The Compro and Pinnacle cards were neck-to-neck in terms of content capture and processor usage. The overall quality of content of both the cards was similar, and we had a tough time choosing between them. Aviosys was another card that performed well. The Pixelview's lost badly—except for the PlayTV XP, which put up a decent show. Some cards required fine-tuning after the automatic channel—the Mercury and Compro yielded grainy pictures, which cleared when we fine-tuned. The Pixelview PlayTV PRO also required fine-tuning, but gave pretty decent content quality. The processor usage remained the same across the board—all the cards utilised 21 per cent processor time, when capturing data.

Overall, the Compro and the Mercury came out as our performance and value winners, but the Pinnacle duo does deserve a worthy mention for their commendable performance. The Compro won because of the features it packed and the performance it showed, while the Mercury won because of its extremely competitive price of just Rs 1,700 that no other card was able to match.

Compro	Mercury	Pinnacle	Pixelview
Mediatech India	Kobian India Ltd	Aditya Infotech P.ltd	Rashi Peripherals
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